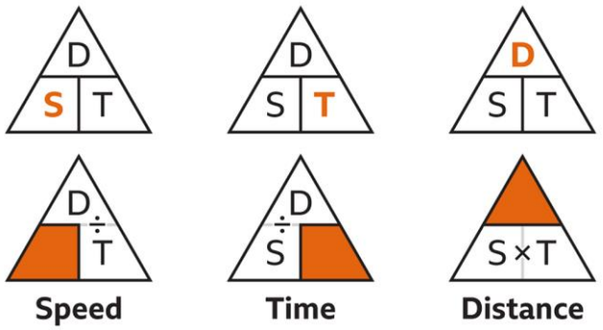
Knowledge Organiser
Year 7: Forces – Speed, distance and time

What is speed?
- Speed is a measure of how fast an object is moving
- It is calculated by dividing the distance travelled by the time taken
- The units of speed depend on the units of distance and time used e.g. metres per second (m/s) or miles per hour (mph)

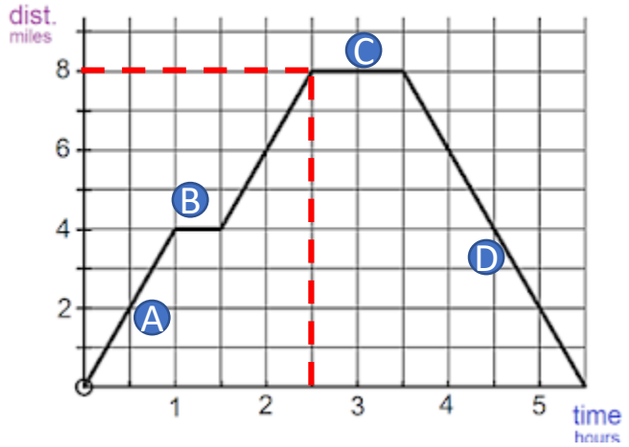
$$\text{SPEED (m/s)} = \frac{\text{DISTANCE (m)}}{\text{TIME (s)}}$$

An equation showing how to calculate speed in m/s from distance and time.



Three formula triangles showing how to rearrange s = d/t to calculate time and distance.

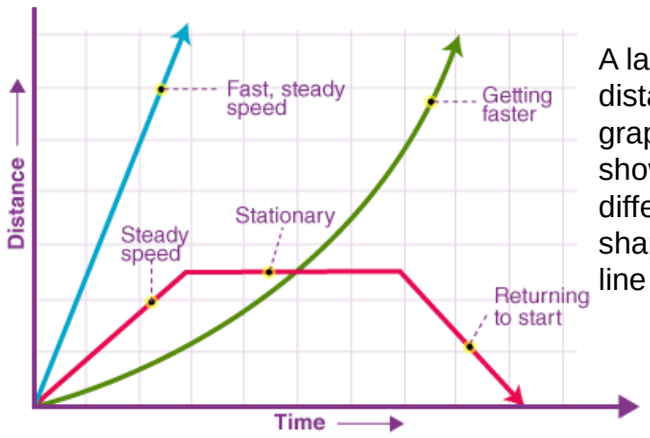
What is a distance-time graph?
- A distance-time graph shows how the distance travelled by an object changes over time
- Distance is plotted on the **x-axis** and time on the **y-axis**
- The steepness of the line represents the speed of the object at that point in time



How do we read a distance time graph?
- If you want to know the **distance travelled from the starting point** at a given time you need to draw a line up from that time
- When your reach the line of the data, go across to the distance and read off
- In the graph above, we can see that after 2.5 hours the car had travelled 8 miles from the starting point

What does the slope of the line mean?
- We say a line has a **steeper slope** if it goes up more quickly
- The steepness of the slope at each point on the graph indicates the speed at which the car is travelled

Point A = The car is moving at a constant speed between 0-1 hours
Point B = The car is stationary (not moving) between 1-1.5 hours
Point C = The car is moving at the same constant speed as during A
Point D = The car is travelling at a constant speed, back in the direction of the start point. We can tell because the line is going down



A labelled distance/time graph showing the different shapes the line can have

What is acceleration?
- Acceleration is the rate at which the speed of an object changes
- It is calculated by dividing the change in speed by the time taken
- The units of acceleration are m/s^2

Quantity	Symbol	Unit	Equation
Speed	s	m/s	s = d/t
Distance	d	m	d = s x t
Time	t	s	t = d/s
Acceleration	a	m/s^2	a = Δs/t

A table which shows the symbol, unit and equation used to calculate speed, distance, time and acceleration

How is acceleration different to speed?
- Acceleration is the change in speed whereas speed is this distance covered per time
- An object can be travelling slowly but have a high acceleration
- It can also be travelling quickly and have a low acceleration

Think about a cheetah and a race car.



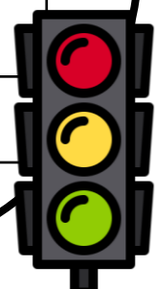
The cheetah has a **higher acceleration** because it can get to it's top speed much quicker.

The race car has a **higher top speed** but takes longer to get there, so has a **lower acceleration.**



RAG Yourself!

Learning Outcomes;	RAG
State the equation that links speed, distance and time.	
State the units of speed, distance and time	
Draw distance/time graphs	
Read and understand the shape of a distance/time graph	
Explain the difference between speed and acceleration	



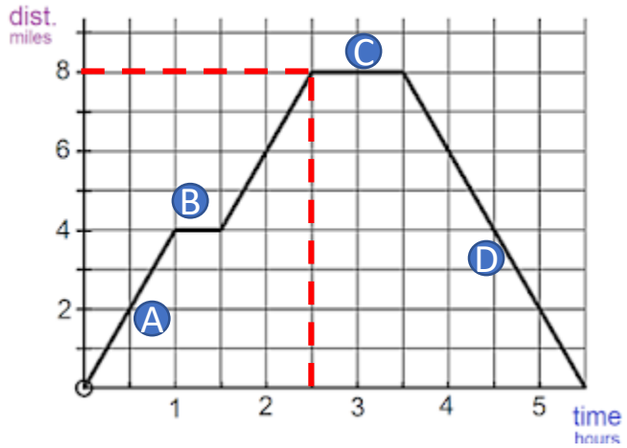
Knowledge Organiser
Year 7: Forces – Speed, distance and time

What is speed?

An equation showing how to calculate speed in m/s from distance and time.

Three formula triangles showing how to rearrange s = d/t to calculate time and distance.

What is a distance-time graph?



How do we read a distance time graph?

What does the slope of the line mean?

Point A =

Point B =

Point C =

Point D =

A labelled distance/time graph showing the different shapes the line can have

What is acceleration?

Quantity	Symbol	Unit	Equation
Speed			
Distance			
Time			
Acceleration			

A table which shows the symbol, unit and equation used to calculate speed, distance, time and acceleration

How is acceleration different to speed?





RAG Yourself!

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